Image-Enhanced Coarse-to-Fine YOLO26n-seg for Real-Time Lymphatic Vessel Localization in ICG Lymphography

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Abstract

Near-infrared indocyanine green (ICG) lymphography is widely used to visualize superficial lymphatic flow during lymphaticovenous anastomosis (LVA). However, dermal reflux, tissue scattering, and low vessel-to-background contrast can make fine lymphatic vessels difficult to localize in individual frames of clinical ICG videos. This work presents an image-enhanced coarse-to-fine pipeline for real-time frame-wise lymphatic vessel segmentation. For each frame, a lightweight YOLO26n detector localizes the foot region of interest, after which median filtering, white top-hat transformation, and percentile contrast stretching are applied to suppress non-uniform fluorescence backgrounds and enhance faint tubular structures. YOLO26n-seg then generates a pixel-level lymphatic vessel mask. A total of 492 diagnostically usable ICG frames selected from videos of 164 patients were used for model development and evaluation. Compared with the original-image baseline, the proposed pipeline improved the Dice coefficient from 0.7318 to 0.9050 and the IoU from 0.6152 to 0.8333. In frame-wise video processing experiments, the average core processing time was 18.39–19.38 ms per frame, and the maximum end-to-end processing throughput was approximately 30 frames per second under unpaced decoding. These results demonstrate accurate segmentation and low processing latency on the selected clinical frames. The current model processes frames independently and does not explicitly model temporal information across adjacent frames.

Keywords: ICG lymphography, lymphatic venous anastomosis, lymphatic vessel segmentation, YOLO26n-seg, white top-hat transform, low-latency frame-wise processing.

1. Introduction

Limb lymphedema is a chronic complication commonly observed following breast and gynecological cancer treatments, where impairment of the lymphatic transport system leads to abnormal accumulation of interstitial fluid, severely degrading patients' quality of life [1, 2]. In clinical practice, LVA has emerged as a primary microsurgical therapeutic approach. Successful LVA depends on accurate preoperative and intraoperative localization of physiologically functional lymphatic vessels. From the perspective of optical imaging, ICG lymphography represents the gold standard for observing dynamic lymph flow and tubular topological structures [3]. However, due to obstructed lymphatic drainage, the fluorescent contrast agent frequently extravasates into surrounding tissues, causing the phenomenon of dermal reflux, which is exceptionally challenging to manage clinically. Accompanied by subcutaneous tissue fibrosis, this phenomenon introduces substantial non-linear, high-frequency optical scattering and noise into the images, precipitously degrading the contrast between fine lymphatic vessels and the background. In current clinical practice, anatomical path planning still relies heavily on clinicians' visual interpretation and manual target delineation on static monitors. This process is time-consuming and highly dependent on operator experience. It can also cause substantial visual fatigue and introduce subjective uncertainty into clinical decision-making.

Advances in computer vision and medical image anal-

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ysis have facilitated the integration of optical imaging with artificial intelligence for automated feature extraction [4]. For the segmentation of thin tubular structures, Liu et al. [5] developed a dual-stage network with pixel emendation, whereas Lv et al. [6] introduced an attention-guided U-Net incorporating atrous convolution. Although these methods have demonstrated accurate retinal vessel segmentation, their direct application to ICG lymphography remains challenging because dermal reflux, tissue scattering, uneven illumination, and diffuse fluorescence can obscure vessel boundaries or produce vessel-like background patterns. More complex architectures, including the multi-task attention network proposed by Ling et al. [7] and the Mamba-based medical image segmentation framework developed by Liu et al. [8], further improve semantic representation and boundary delineation. However, their computational requirements may limit their practicality in latency-sensitive intraoperative environments [9]. In addition, many existing approaches focus primarily on offline analysis of individual images rather than low-latency processing of consecutive ICG video frames [10]. The present study does not explicitly model temporal dependencies between frames; instead, it applies independent frame-wise segmentation to ICG video sequences. To address the image-quality challenges within each frame, a fixed preprocessing pipeline comprising median filtering, a morphological white top-hat transformation, and percentile-based contrast stretching is applied before segmentation. These operations are intended to reduce diffuse background variation and enhance faint tubular structures, thereby providing more distinguishable inputs for the subsequent lightweight segmentation model.

Compared with offline static medical image analysis performed on server-class hardware, intraoperative ICG video processing must operate under tighter computational constraints and variable imaging conditions [11, 12]. Individual video frames may contain substantial non-diagnostic regions, uneven illumination, background fluorescence, motion artifacts, and surgical instruments. Applying a segmentation model directly to the full frame may increase the computational burden and expose the model to irrelevant background features [13]. Therefore, the proposed pipeline separates coarse ROI localization from local pixel-level vessel segmentation. This design is intended to restrict the segmentation input to the clinically relevant anatomical region and reduce background interference. To quantify the contribution of ROI localization, the ROI-based configuration was compared with direct full-frame segmentation using the same YOLO26n-seg architecture and evaluation metrics. The comparison focuses on segmentation accuracy, whereas the independent computational contribution of ROI localization was not assessed. Based on these clinical and engineering considerations, this study presents a clinically motivated coarse-to-fine processing pipeline for low-latency, frame-wise lymphatic vessel segmentation in ICG video sequences [14]. The pipeline combines per-frame ROI localization, fixed image-enhancement operations, and a lightweight instance-segmentation model. Each frame is processed independently, and the predicted vessel masks are subsequently mapped back to the original frame coordinates for sequential intraoperative visualization. Accordingly, the proposed method should be interpreted as a hybrid frame-processing pipeline rather than a jointly trainable end-to-end or temporally aware video-segmentation architecture.

The main contributions of this work are summarized as follows:

- A lightweight YOLO26n detector is used for per-frame localization of the affected foot region, with symmetric zero padding applied to standardize the input size without distorting anatomical structures.
- A three-stage enhancement pipeline combining median filtering, white top-hat transformation, and percentile-based contrast stretching is used to reduce background variation and enhance faint lymphatic vessels.
- YOLO26n-seg is employed for pixel-level segmentation of lymphatic vessels within the localized ROI.
- The predicted masks are mapped back onto the original video frames to provide frame-wise visualization for potential intraoperative applications.
- Comparative experiments demonstrate a favorable balance between segmentation accuracy and computational efficiency under the evaluated conditions.

The remainder of this paper is organized as follows. Section 2 presents a review of recent advances in ICG lymphography, computer-aided lymphatic image analysis, and deep learning-based vessel segmentation for LVA-related applications. Section 3 describes the proposed image-enhanced, coarse-to-fine YOLO26n-seg pipeline, including per-frame ROI localization, aspect-ratio-preserving input standardization, the three-stage image-preprocessing procedure, and pixel-level lymphatic vessel segmentation. Section 4 describes the clinical dataset, ground-truth annotation procedure, evaluation metrics, implementation environment, and experimental configuration. Section 5 reports the quantitative results, including the preprocessing ablation study, segmentation-model comparison, validation performance, and computational-efficiency assessment. Section 6 discusses the clinical implications, observed failure cases, and limitations of the proposed pipeline. Finally, Section 7 concludes the paper and outlines directions for future research on low-latency lymphatic vessel segmentation in ICG video sequences.

2. Related Work

2.1. Clinical Background and Challenges in ICG Lymphography

In the clinical workflow of LVA, preoperative and intra-

operative localization of functional lymphatic vessels has gradually shifted from empirical anatomical estimation to ICG lymphography, which enables real-time visualization of superficial lymphatic drainage pathways. For instance, Hara and Mihara [15] applied multi-lymphosome injection ICG lymphography by injecting ICG into three lower-limb lymphosomes to improve the detection of functional lymphatic vessels for lymphedema surgery. Their study included 206 lower limbs from 103 patients and reported visible linear lymphatic patterns in 84.5% of saphenous lymphatics, 72.8% of lateral calf lymphatics, and 75.2% of lateral thigh lymphatics. Notably, at least one linear pattern was detected in 83.3% of limbs classified as lymphoscintigraphic Type 5. However, this approach remains dependent on visible fluorescence patterns and clinical interpretation, and it does not provide pixel-level lymphatic vessel masks for direct surgical overlay.

Akita et al. [16] further validated the clinical utility of ICG fluorescent lymphography in a phase III multicenter study for secondary lymphedema assessment and LVA planning. Their study reported diagnostic accuracy of 100.0% in 26 upper extremities and 88.2% in 88 lower extremities, and lymphatic vessels suitable for anastomosis were identified at 97.7% of 263 incision sites. Nevertheless, ICG lymphography remains highly dependent on visual interpretation by experienced surgeons, especially when dermal reflux, tissue scattering, and non-uniform fluorescence obscure fine lymphatic vessels.

2.2. Automated Analysis of Lymphatic Imaging

To reduce manual workload in lymphedema image interpretation, Nomura et al. [17] developed a computer-aided diagnosis system for lower-extremity lymphedema screening using pelvic computed tomography images and deep convolutional neural networks. Their study used 431 pelvic CT images from 154 gynecological cancer patients, with ICG dermal backflow staging used as the clinical reference, and the ResNet-34 [18] model with fat-enhanced images achieved an AUC of 0.967 and an accuracy of 92.9%. Nevertheless, this method performs image-level lymphedema screening from CT images rather than real-time lymphatic vessel segmentation in ICG lymphography videos, limiting its direct use for intraoperative LVA guidance.

On the other hand, Tai et al. [19] developed LVADet, a computer-vision-based system for lymphatic vessel detection in ICG images, using Attention U-Net [20] for lymphatic vessel segmentation, keypoint R-CNN with ResNet-50-FPN [21] for foot landmark detection, and Procrustes analysis [22] for mirror mapping from the unaffected limb to the affected limb. The system was evaluated using 52 lymphedema cases, including 42 training cases and 10 testing cases, and the segmentation module achieved an accuracy of 0.9851, AUC of 0.9909, sensitivity of 0.7705, and specificity of 0.9897. Although LVADet provides an important step toward computer-assisted LVA planning, its localization relies on bilateral geometric mapping and may be affected when the affected limb presents severe dermal reflux, asymmetric anatomy, or weak visible lymphatic pathways.

Recently, Ozmen et al. [23] introduced an AI-based ICG lymphography pattern classification framework using a MobileNetV2 [24] network architecture to categorize images into linear, reticular, splash, stardust, and diffuse patterns. Their model achieved 97.78% accuracy on a dataset of 68 ICG lymphography images, indicating the potential of deep learning for standardizing ICG image interpretation. However, the task was formulated as image-level pattern classification rather than pixel-level lymphatic vessel segmentation, limiting its ability to provide spatial vessel masks for intraoperative guidance.

2.3. Deep Learning-Based Vessel Segmentation

In biomedical image segmentation, encoder–decoder networks and topology-aware methods have been widely applied to thin tubular structures such as blood vessels, neurons, and lymphatic vessels. For example, Shit et al. [25] proposed clDice, a topology-preserving metric and loss function for tubular structure segmentation, and validated it using multiple two-dimensional and three-dimensional datasets. Their results demonstrated that centerline-based evaluation can represent structural continuity more effectively than overlap-based metrics alone. However, topology-aware losses and metrics primarily address structural consistency and do not directly resolve the low contrast, diffuse fluorescence background, or computational constraints encountered in intraoperative ICG lymphography.

Lightweight one-stage instance-segmentation models provide a computationally efficient approach for frame-wise medical image analysis because object localization and mask prediction can be performed within a single model pass. YOLO26 [26] is designed as a lightweight real-time vision architecture with instance-segmentation capability. Nevertheless, generic segmentation models are not specifically designed for ICG lymphography, in which lymphatic vessels may appear faint, slender, discontinuous, and partially obscured by dermal reflux or uneven background fluorescence.

Temporal video-segmentation methods can further exploit relationships between consecutive frames to improve prediction consistency [11, 12]. However, the present study does not incorporate optical flow, recurrent networks, temporal attention, or other inter-frame feature-fusion mechanisms. Instead, it combines per-frame ROI localization, fixed image-enhancement operations, and lightweight instance segmentation. Therefore, the proposed method should be regarded as a frame-wise segmentation pipeline applied to ICG video sequences rather than a temporally aware video-segmentation model.

2.4. Research Gap

Although previous studies have demonstrated the clinical value of ICG lymphography and the feasibility of

computer-aided lymphatic image analysis, they have generally addressed different clinical or computational tasks. Clinical ICG lymphography techniques improve the visualization of functional lymphatic pathways, but interpretation remains dependent on clinical expertise, and automated pixel-level masks are not routinely provided for intraoperative overlay [15, 16]. CT-based computer-aided screening can identify lower-extremity lymphedema with high diagnostic performance but was not designed for lymphatic vessel segmentation in ICG images [17]. Existing LVA planning systems can localize lymphatic vessels and anatomical landmarks; however, approaches based on bilateral geometric mapping may be less reliable in patients with substantial anatomical asymmetry or severe dermal reflux [19]. Image-level classification models can standardize the recognition of ICG lymphography patterns but do not delineate vessel morphology or generate spatial masks [23]. Similarly, topology-aware methods improve the evaluation and preservation of tubular continuity but do not directly address fluorescence degradation or computational efficiency [25]. Generic lightweight segmentation architectures support efficient deployment but are not specifically designed for faint lymphatic vessels affected by tissue scattering, dermal reflux, and nonuniform fluorescence [26].

Taken together, these studies indicate a need for a lightweight processing pipeline that can reduce background interference, enhance faint vessel structures, and generate pixel-level masks from ICG video frames with low processing latency. The present study addresses this practical gap through a coarse-to-fine, frame-wise strategy. A lightweight detector first localizes the clinically relevant foot region, followed by aspect-ratio-preserving input standardization and fixed image-enhancement operations. A lightweight segmentation model then generates lymphatic vessel masks, which are mapped back to the original frame coordinates for visualization. Each frame is processed independently without temporal feature fusion. Therefore, the proposed method should be regarded as an integrated frame-processing pipeline rather than a temporally aware or jointly trainable end-to-end architecture. A direct full-frame segmentation baseline was included to quantify the contribution of ROI localization to segmentation performance.

3. Proposed Method

This work presents a low-latency, frame-wise processing pipeline for lymphatic vessel segmentation in near-infrared ICG video sequences. For each input frame, the pipeline performs ROI localization, spatial standardization, image enhancement, lightweight instance segmentation, and mask remapping. The predicted vessel masks are then overlaid on the corresponding original frames for sequential visualization, as illustrated in Figure 1.

The processing pipeline begins with per-frame foot-region localization using YOLO26n. The detected ROI is standardized using aspect-ratio-preserving scaling and symmetric zero padding. A fixed three-stage enhancement procedure comprising median filtering, white top-hat transformation, and percentile-based contrast stretching is subsequently applied. YOLO26n-seg then generates a pixel-level lymphatic vessel mask within the processed ROI, which is mapped back to the original frame coordinates.

Each frame is processed independently. The pipeline does not use optical flow, recurrent connections, temporal attention, temporal memory, or inter-frame feature fusion. Therefore, the proposed method performs frame-wise segmentation on ICG video sequences rather than temporally aware video segmentation.

Table 1: Clinical characteristics of the dataset

Characteristic	Category	# patients
Dataset size	Total patients	164
Anatomical region	Lower limb	164
Sex	Male	54
	Female	110
Etiology	Malignant cancer	122
	Injury/trauma	12
	Primary lymphedema	19
	Others	11

3.1. Clinical Dataset Acquisition and Curation

The ICG lymphography data utilized in this work were retrospectively collected at Taichung Veterans General Hospital, Taiwan. All subjects included in the analysis were clinical patients undergoing preoperative and intraoperative lymphatic vessel mapping for LVA. The data collection protocol received formal approval from the Institutional Review Board (IRB) of the hospital (Approval No.: CMUH113-REC3-187), and written informed consent for clinical data usage was obtained from all patients.

The clinical characteristics of the study cohort are summarized in Table 1. The cohort comprised 164 patients, with all ICG lymphography data acquired from the lower limbs. Of these patients, 54 were male and 110 were female. With respect to lymphedema etiology, malignancy-related cases constituted the largest subgroup, accounting for 122 patients. The remaining cohort included 12 patients with injury- or trauma-related lymphedema, 19 with primary lymphedema, and 11 with other etiologies.

The following subsections describe the acquisition and curation of the clinical dataset, per-frame ROI localization and standardization, the image preprocessing pipeline, and the instance segmentation architecture.

The ICG lymphography acquisition protocol is summarized in Table 2. All images were acquired using the FLUOBEAM LM imaging system (FLUOPTICS SAS, Grenoble, France). For contrast preparation, a 25 mg vial of indocyanine green was reconstituted with 10 mL of sterile water to obtain a final concentration of 2.5 mg/mL.

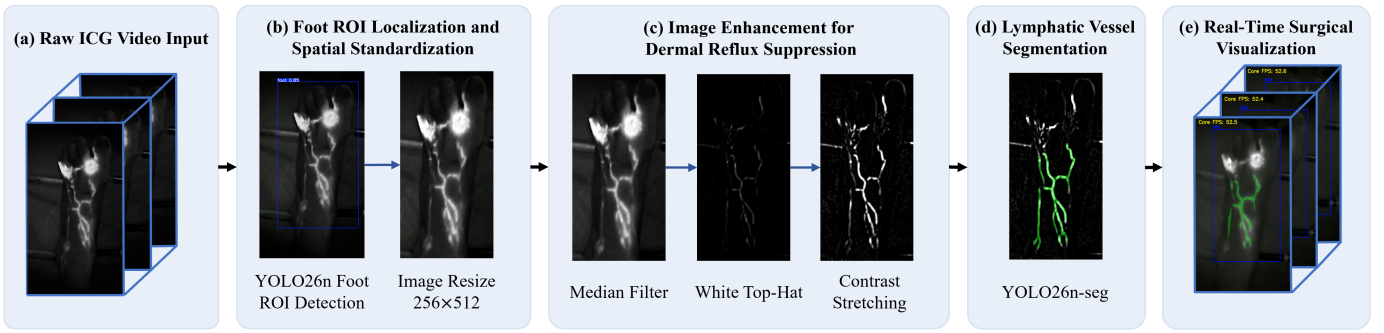


Figure 1: Overall workflow of the proposed frame-wise pipeline for lymphatic vessel segmentation and mask visualization.

Table 2: Summary of the ICG lymphography acquisition protocol

Item	Description
Imaging system	FLUOBEAM LM (FLUOPTICS SAS, Grenoble, France)
ICG preparation	Indocyanine green 25 mg/vial diluted with 10 mL sterile water
ICG concentration	2.5 mg/mL
Injection sites	First and fourth interdigital spaces of the affected lower limb
Injection route	Subcutaneous injection
Acquisition timing	Approximately 1 min after injection

The prepared ICG solution was administered subcutaneously into the first and fourth interdigital spaces of the affected lower limb. Image acquisition commenced approximately 1 min after injection to capture the early lymphatic drainage phase.

The original dataset consisted of intraoperative ICG lymphography video sequences. To construct an image-based dataset suitable for YOLO-based detection and segmentation, three diagnostically usable and reliably annotatable frames were manually selected from each patient by an experienced microsurgeon, resulting in 492 frames from 164 patients. Frame selection was based on the visibility of the affected foot region and the ability to identify the principal lymphatic vessel structures with sufficient confidence for pixel-level annotation, rather than on fixed time points. The selected frames included variations in vessel contrast, dermal reflux, tissue scattering, and background fluorescence. Frames with severe motion blur, incomplete anatomical coverage, or vessel boundaries that could not be annotated reliably were excluded.

For model development and evaluation, the dataset was divided using a patient-level 8:2 splitting strategy to avoid data leakage between the training and testing subsets. The training subset comprised 131 patients and 393 frames, representing 79.9% of the complete dataset. The testing subset comprised the remaining 33 patients and 99 frames, corresponding to 20.1%. This patient-level partitioning ensured that frames from the same patient were not distributed across both subsets. Consequently, the risk of patient-specific information leakage was minimized. The resulting dataset distribution is summarized in Table 3. For the establishment of reliable ground-truth annotations, all 492 frames were manually labeled with pixel-level lymphatic vessel masks using Roboflow [27]. The vessel boundaries were delineated and verified by one microsurgeon with six years of experience in LVA. Ambiguous vessel boundaries were carefully reviewed by the same microsurgeon before finalization. These clinician-reviewed annotations served as the reference masks for model training and quantitative performance evaluation.

Table 3: Patient-level dataset split

Subset	Patients	Frames	Ratio
Training	131	393	79.9%
Testing	33	99	20.1%
Total	164	492	100%

3.2. Per-frame ROI Localization and Standardization

Individual clinical ICG frames may contain substantial non-diagnostic regions, background fluorescence, surgical instruments, and other irrelevant visual information. To restrict the subsequent segmentation analysis to the clinically relevant anatomical region, YOLO26n [26] is used to independently localize the affected foot in each frame. The detector outputs a two-dimensional bounding box that defines the ROI for subsequent cropping, standardization, image enhancement, and vessel segmentation. No inter-frame tracking or temporal state propagation is applied.

Following ROI localization, the detected foot region is extracted from each video frame through pixel-level axis-aligned cropping. Owing to variations in camera angle, working distance, and patient anatomy, the cropped ROIs exhibit considerable differences in spatial dimensions. Direct resizing to a fixed input resolution may introduce geometric deformation and compromise the morphological integrity of fine lymphatic vessels. To ensure dimensional consistency for the downstream instance segmentation network, an aspect-ratio-preserving standardization scheme is therefore employed.

Specifically, each cropped ROI is isotropically rescaled until its width reaches 256 pixels, while the original aspect ratio is maintained. The resulting height is then adjusted to the target dimension of 512 pixels through symmetric zero padding. Zero-valued pixels are distributed evenly along the upper and lower image boundaries. This procedure produces a standardized input tensor of 256×512 pixels while preserving the anatomical proportions of the foot and the structural characteristics of the lymphatic vessels. In contrast to conventional image stretching, the proposed strategy minimizes geometric distortion and provides spatially consistent inputs for subsequent segmentation. The complete ROI standardization procedure is illustrated in Figure 2.

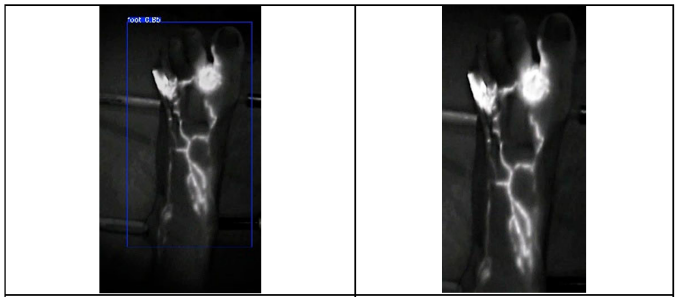


Figure 2: Schematic of image segmentation: (a) localized foot region; (b) image standardization.

3.3. Image Preprocessing Pipeline

In ICG lymphography, fluorescence from subcutaneous tissue is often affected by light scattering, tissue fibrosis, and dermal reflux. These effects produce diffuse background signals and irregular high-intensity regions, which reduce the contrast between lymphatic vessels and surrounding tissue. In severe cases, the intensity distributions of vessels and background regions may overlap, making vessel delineation more difficult [28].

To improve image consistency before segmentation, a fixed three-stage preprocessing pipeline is implemented. The pipeline combines established image-processing operations, including median filtering, morphological white top-hat transformation, and percentile-based contrast stretching. These operations are fixed and are not jointly optimized with the segmentation network. As shown in Figure 3, the steps are mainly to remove noise, correct background variation, and enhance vessel visibility. Particularly, median filtering is first applied to suppress isolated intensity fluctuations caused by sensor noise and fluorescence irregularities. Unlike conventional averaging filters, it reduces impulsive noise without excessively smoothing narrow vessel boundaries. A WTH transformation is then used to separate small bright structures from the slowly varying background. In this operation, the background component is estimated through morphological opening and removed from the original image. As a result, faint

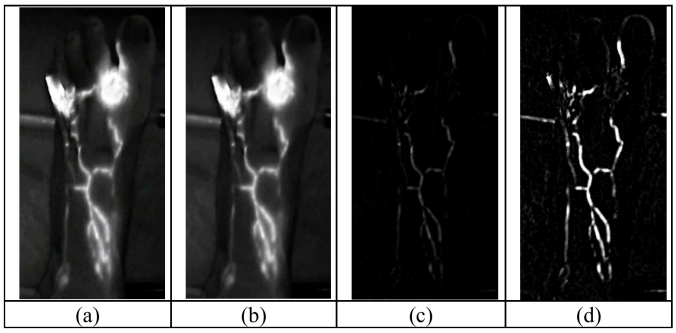


Figure 3: Illustration of the image processing steps: (a) ROI image; (b) median filtering; (c) white top-hat transformation; (d) contrast stretching.

tubular structures become more distinguishable, whereas broad regions of diffuse fluorescence are attenuated. Finally, percentile-based contrast stretching expands the intensity range of the enhanced vessel responses. The utilization of percentile limits reduces the influence of extreme intensity outliers and prevents a small number of saturated pixels from dominating the contrast adjustment. The individual stages of this image-processing pipeline are described below.

A. Non-linear Spatial Median Filtering

During near-infrared-guided dynamic ICG lymphography acquisition, raw image sequences are frequently corrupted by severe discrete impulse noise. This degradation primarily stems from sensor thermal noise in low-light imaging environments and the random scattering of unbound free fluorescent molecules within subcutaneous tissues. To suppress localized background noise while preserving the sharp boundaries and spatial continuity of fine lymphatic vessels, a 3×3 nonlinear median filter is applied during image preprocessing. The filtering operation is defined as

$$I_{\text{med}}(x, y) = \text{median}_{i,j \in \{-1,0,1\}} I(x + i, y + j) \qquad (1)$$

where $I(x, y)$ denotes the intensity value of the input image at pixel location (x, y), and $I_{\text{med}}(x, y)$ represents the corresponding filtered intensity. The variables i and j denote the horizontal and vertical offsets within the 3×3 neighborhood centered at (x, y). The median operator replaces the central pixel with the median of the nine intensity values in this local neighborhood.

Notably, conventional linear smoothing filters, such as Gaussian filters, can attenuate noise but may also blur anatomical boundaries and weaken fine vessel structures [29]. To mitigate this limitation, a nonlinear median filter is adopted as an order-statistics-based operator. By replacing each pixel with the median intensity within its local neighborhood, the filter effectively suppresses isolated high-intensity outliers and anomalous fluorescent spots. A 3×3 window is employed to achieve a suitable balance between noise reduction and structural preservation. This

compact neighborhood limits excessive smoothing while maintaining the continuity and edge definition of narrow tubular vessels. Consequently, the median-filtered image provides a cleaner and more stable input for the subsequent morphological enhancement stage.

B. Morphological White Top-Hat Transform

When processing ICG images of advanced lymphedema, fluorescence leakage induced by dermal reflux frequently exhibits a high degree of spatial non-uniformity, forming large-scale and low-frequency optical diffuse backgrounds. Conventional linear high-pass filters or adaptive thresholding techniques are highly susceptible to edge ringing artifacts or excessive amplification of residual discrete noise when coping with such non-linear and gradient illumination interference. To reduce the influence of slowly varying background fluorescence, a morphological white top-hat transformation [30] is applied as the second preprocessing stage.

The WTH transformation estimates and removes slowly varying background fluorescence while preserving small bright structures. Since lymphatic vessels generally occupy a smaller spatial scale than the diffuse fluorescence patterns caused by tissue scattering and dermal reflux, the transformation can distinguish fine vessel responses from the non-uniform background. This enhances vessel visibility and preserves spatial organization.The background component is estimated through morphological opening, which consists of erosion followed by dilation. This estimated background is then subtracted from the median-filtered image. The WTH transformation is defined as

$$I_{\mathrm{WTH}}(x,y) = I_{\mathrm{med}}(x,y) - (I_{\mathrm{med}} \circ S)(x,y) \qquad (2)$$

where $I_{\mathrm{med}}(x,y)$ denotes the median-filtered intensity at pixel location (x,y), and $I_{\mathrm{WTH}}(x,y)$ represents the corresponding WTH response. The symbol $\circ$ denotes the morphological opening operation with structuring element S, defined as

$$I_{\mathrm{med}} \circ S = (I_{\mathrm{med}} \ominus S) \oplus S, \qquad (3)$$

where $\ominus$ and $\oplus$ denote morphological erosion and dilation, respectively. The structuring element S determines the spatial scale and geometric characteristics of the structures retained by the transformation. In this work, S is implemented as a 9×9 disk-shaped structuring element. The disk geometry provides an approximately isotropic response and reduces orientation-dependent enhancement. This property is particularly important because lymphatic vessels may appear along different anatomical directions. The selected spatial support is larger than the expected vessel width but smaller than the broad fluorescence regions associated with dermal reflux. Consequently, the opening operation primarily captures the slowly varying background, whereas the subtraction step preserves fine bright tubular structures. The 9×9 configuration also provides a practical balance between vessel enhancement and computational efficiency.

By attenuating slowly varying background responses, the WTH transformation is intended to improve the visibility of faint lymphatic vessel structures. It also reduces large-scale background responses, thereby providing a cleaner input with an improved signal-to-noise ratio for subsequent processing.

C. Percentile Contrast Stretching

Although the WTH transformation suppresses slowly varying background fluorescence, the resulting vessel responses may remain confined to a narrow intensity range. Consequently, faint lymphatic structures may exhibit limited contrast relative to the residual background. Directly using these low-amplitude responses may reduce their separability in the subsequent segmentation network. To improve vessel saliency, percentile-based contrast stretching is applied as the final preprocessing stage.

The contrast adjustment is defined as

$$I_{\mathrm{out}}(x,y) = \mathrm{clip}_{[0,255]}\left[\frac{I_{\mathrm{WTH}}(x,y) - P_1}{P_{99} - P_1} \times 255\right], \qquad (4)$$

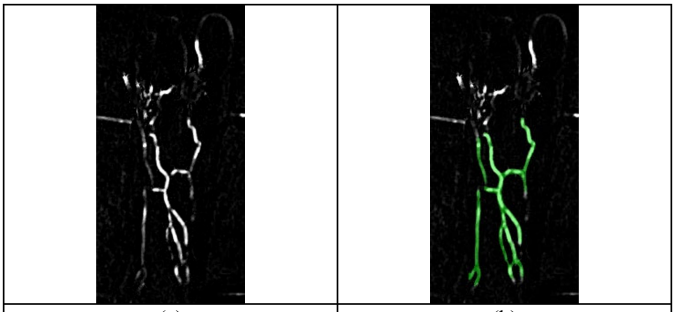
where (x,y) denotes the spatial coordinates of a pixel. The term $I_{\mathrm{WTH}}(x,y)$ represents the pixel intensity after WTH transformation, whereas $I_{\mathrm{out}}(x,y)$ denotes the contrast-enhanced output intensity. The parameters P_1 and P_{99} are the first and 99th percentiles of the grayscale intensity distribution of I_{WTH}, respectively. The operator $\mathrm{clip}_{[0,255]}(\cdot)$ constrains the transformed intensity values to the valid 8-bit range. Accordingly, values below 0 are assigned to 0, whereas values above 255 are assigned to 255.

Compared with conventional min–max normalization, percentile-based stretching is less sensitive to isolated saturated pixels and extreme intensity outliers. The lower and upper percentile bounds suppress the influence of atypical responses while expanding the intensity range occupied by faint vessel structures. This procedure improves vessel-to-background contrast and reduces intensity variations caused by differences in imaging conditions, fluorescence distribution, and patient-specific tissue characteristics. Importantly, the transformation modifies only the intensity distribution and therefore preserves the spatial organization of the lymphatic vessels.

3.4. Instance Segmentation Architecture

Following image preprocessing, YOLO26n-seg [26] is adopted as the lightweight instance-segmentation component.Low-latency frame-wise visualization requires both efficient inference and accurate delineation of fine lymphatic vessels [31, 32]. The lightweight network was therefore selected to balance segmentation accuracy and computational efficiency when processing individual frames extracted from ICG video sequences.

For each video frame t, the enhanced ROI image is denoted by R_t^{enh}. The segmentation model performs multi-scale feature extraction, feature fusion, and mask decoding



(a) (b)

Figure 4: Image segmentation results: (a) preprocessed image; (b) lymphatic segmentation map.

to estimate the spatial distribution of lymphatic vessels. This mapping is expressed as

$$P_t = F_\Theta\left(R_t^{\text{enh}}\right), \tag{5}$$

where $F_\Theta(\cdot)$ denotes the YOLO26n-seg model parameterized by the learnable parameters Θ. The input $R_t^{\text{enh}} \in \mathbb{R}^{H\times W\times C}$ represents the preprocessed image at frame t, where H, W, and C denote the image height, width, and number of channels, respectively. The output $P_t \in [0,1]^{H\times W}$ denotes the vessel probability map after mask decoding. Each element $P_t(x,y)$ indicates the predicted probability that the pixel at spatial location (x,y) belongs to the lymphatic vessel class.

A thresholding operation is subsequently applied to convert the continuous probability map into a binary vessel mask:

$$M_t(x,y) = \begin{cases} 1, & P_t(x,y) \geq \tau_s, \\ 0, & P_t(x,y) < \tau_s, \end{cases} \tag{6}$$

where $M_t(x,y) \in \{0,1\}$ denotes the binary segmentation result at pixel location (x,y), and $\tau_s \in [0,1]$ represents the segmentation threshold. Pixels with probabilities equal to or greater than τ_s are classified as lymphatic vessels, whereas the remaining pixels are assigned to the background. This decision rule suppresses low-confidence responses and produces explicit pixel-level vessel boundaries.

As illustrated in Figure 4, the resulting masks preserve the continuity and slender morphology of lymphatic vessels under heterogeneous fluorescence conditions. The binary masks are subsequently mapped to the corresponding original video frames and displayed as color-coded overlays. This frame-by-frame visualization provides interpretable spatial feedback and supports real-time identification of candidate regions for LVA.

4. Experimental Setup

This section presents the experimental setup used to evaluate the proposed ICG lymphatic vessel segmentation framework. Specifically, the hardware and software configurations, training settings, and hyperparameter selections are introduced to ensure experimental reproducibility. Then, the performance metrics used for ROI localization, lymphatic vessel segmentation, and real-time video inference are defined and elaborated.

Table 4: Hardware and software configurations

Component	Specification/ Version
Operating system	Microsoft Windows 11
GPU	NVIDIA GeForce RTX 5060
Programming language	Python 3.10
Deep learning framework	PyTorch 2.12.0 / CUDA 12.8
Image analytics tool	OpenCV 4.8

Table 5: Summary of hyperparameter settings for model training process

Parameter	Value/ Setting
Optimizer	AdamW
Initial learning rate	1×10^{-4}
Weight decay	0.05
Learning rate scheduler	Cosine Annealing
Loss function	Label smoothing cross-entropy
Smoothing factor	0.1
Batch size	32
Epoch	100

4.1. Implementation Environment and Training Configuration

All experiments were conducted on a local workstation configured with Microsoft Windows 11. The hardware environment was equipped with an NVIDIA GeForce RTX 5060 GPU to support deep learning model training and inference. The software stack was implemented using Python 3.10, with PyTorch 2.12.0 and CUDA 12.8 employed as the primary deep learning framework and GPU acceleration backend. OpenCV 4.8 was used for ROI standardization, image preprocessing, mask remapping, and video overlay generation. The detailed hardware and software configurations are summarized in Table 4.

The hyperparameter settings used to train the proposed model are summarized in Table 5. For instance, the training configuration employed the AdamW optimizer with an initial learning rate of 1×10^{-4} and a weight decay of 0.05. A cosine annealing learning-rate scheduler was adopted to gradually adjust the learning rate during training and improve convergence stability. The model was trained for 100 epochs with a batch size of 32. For YOLO26n-seg, the optimization objective followed the standard segmentation training losses of the YOLO framework, including localization, classification/ objectness, and mask-related loss components.

4.2. Evaluation Protocol and Performance Metrics

To quantitatively evaluate the performance of the proposed framework, standard metrics commonly used in medical image analysis were adopted. For ROI localization, precision, recall, mAP@50, and mAP@50:95 were

used to evaluate whether the detected foot region sufficiently covered the clinically relevant field of view. For lymphatic vessel segmentation, the Dice coefficient and Intersection over Union (IoU) were used to measure the spatial overlap between the predicted vessel mask and the microsurgeon-annotated ground-truth mask. In addition, pixel-level precision, pixel-level recall, and clDice were adopted to further evaluate false-positive vessel predictions, missed vessel pixels, and centerline-based structural continuity.

A. ROI Localization Metrics

For ROI localization, true positives, false positives, and false negatives were determined according to the overlap between the predicted bounding boxes and the annotated foot-region bounding boxes. Precision measures the reliability of predicted ROI boxes, whereas recall measures the ability of the detector to identify all clinically relevant foot regions. The mAP@50 metric evaluates the mean average precision at an IoU threshold of 0.50, while mAP@50:95 reports the averaged mAP over IoU thresholds from 0.50 to 0.95. These metrics were used to quantify the localization performance of YOLO26n before downstream lymphatic vessel segmentation.

B. Lymphatic Vessel Segmentation Metrics

For lymphatic vessel segmentation, the model predicts whether each pixel belongs to the lymphatic vessel region or the background. Let M denote the predicted lymphatic vessel mask and G denote the ground-truth mask annotated by the microsurgeon. The Dice coefficient and IoU were used to evaluate the overlap between M and G. These two metrics are mathematically expressed as follows:

$$\mathrm{Dice}(M, G) = \frac{2|M \cap G|}{|M| + |G|} \tag{7}$$

$$\mathrm{IoU}(M, G) = \frac{|M \cap G|}{|M \cup G|} \tag{8}$$

A higher Dice or IoU value indicates better agreement between the predicted lymphatic vessel region and the clinician-reviewed annotation. While Dice reflects the overall overlap, IoU provides a stricter evaluation by penalizing both over-segmentation and under-segmentation.

C. Pixel-Level and Structural Continuity Metrics

To further analyze segmentation errors, pixel-level precision and recall were computed. Pixel-level precision reflects the proportion of predicted vessel pixels that are true vessel pixels, whereas pixel-level recall reflects the ability of the model to recover all annotated vessel pixels. These metrics are defined as follows:

$$\mathrm{Precision} = \frac{TP}{TP + FP} \tag{9}$$

$$\mathrm{Recall} = \frac{TP}{TP + FN} \tag{10}$$

Since lymphatic vessels are thin, branching, and topologically continuous structures, the centerline Dice coefficient (clDice) was additionally adopted to evaluate the structural consistency between the predicted and ground-truth vessel masks. Let M_p and M_g denote the predicted and ground-truth binary masks, respectively, and let $\mathcal{S}(\cdot)$ represent the morphological skeletonization operator. The topology precision and topology sensitivity are defined as:

$$T_{\mathrm{prec}} = \frac{|\mathcal{S}(M_p) \cap M_g|}{|\mathcal{S}(M_p)| + \epsilon} \tag{11}$$

$$T_{\mathrm{sens}} = \frac{|\mathcal{S}(M_g) \cap M_p|}{|\mathcal{S}(M_g)| + \epsilon}, \tag{12}$$

where T_{prec} measures the proportion of the predicted centerline contained within the ground-truth mask, whereas T_{sens} measures the proportion of the ground-truth centerline recovered by the predicted mask. The clDice score is subsequently calculated as:

$$\mathrm{clDice} = \frac{2T_{\mathrm{prec}}T_{\mathrm{sens}}}{T_{\mathrm{prec}} + T_{\mathrm{sens}} + \epsilon}, \tag{13}$$

where $|\cdot|$ denotes the number of foreground pixels and ϵ is a small positive constant introduced to avoid division by zero. A higher clDice score indicates better preservation of vessel continuity and branching topology.

4.3. Computational Efficiency Assessment

To assess the feasibility of real-time intraoperative deployment, the computational efficiency of the complete framework was evaluated using the test ICG lymphography videos, as summarized in Table 6. For each frame, the execution time was measured for ROI localization, spatial standardization, image preprocessing, YOLO26n-seg vessel segmentation, mask remapping, overlay generation, and video input/output. The reported values represent the mean processing time over all evaluated video frames.

The proposed YOLO26n-seg model required 3.5 ms for model inference and contained 2.6 million parameters. When ROI localization, spatial standardization, image preprocessing, and segmentation inference were jointly considered, the average core processing time was 19.1 ms per frame, corresponding to approximately 52.5 frames per second. After additionally including video decoding, frame acquisition, mask remapping, overlay generation, and output encoding, the complete pipeline achieved a maximum unpaced processing throughput of 30.4 frames per second. By reporting the complete processing pipeline rather than model inference alone, this evaluation provides a more realistic estimate of its suitability for continuous intraoperative ICG lymphatic vessel visualization.

Table 6: Computational efficiency of the proposed framework.

Item	Value
Input video resolution	540×960 pixels
YOLO26n-seg inference time	3.5 ms/frame
Model parameters	2.6 M
Average core processing time	19.1 ms/frame
Core processing throughput	52.5 FPS
Max. unpaced throughput	30.4 FPS

Table 7: ROI detector performance for foot region localization.

Model	Precision	Recall	mAP@50	mAP@50:95
YOLOv8n [33]	0.985	0.909	0.915	0.720
YOLO11n [34]	1.000	0.949	0.989	0.794
YOLO26n [26]	0.998	1.000	0.995	0.755

5. Results

This section presents a comprehensive evaluation of the proposed framework from four complementary perspectives. First, the effectiveness of the image enhancement pipeline is examined through qualitative comparison and ablation analysis. Next, YOLO26n-seg is compared with representative lightweight segmentation models in terms of accuracy, model complexity, and inference speed. The segmentation behavior is then assessed visually by comparing the predicted masks with clinician annotated references. Finally, the real-time video inference performance is evaluated to determine the feasibility of continuous intraoperative deployment.

5.1. ROI Localization Performance

Before lymphatic vessel segmentation, the ROI detector was quantitatively evaluated for its ability to localize the affected foot in intraoperative ICG video frames. Reliable ROI localization is a critical prerequisite for the proposed coarse-to-fine framework, as bounding box errors may exclude diagnostically relevant regions and propagate to the downstream segmentation stage. As summarized in Table 7, YOLOv8n [33], YOLO11n [34], and YOLO26n [26] were compared using precision, recall, mAP@50, and mAP@50:95.

YOLO26n achieved a precision of 0.998, a recall of 1.000, an mAP@50 of 0.995, and an mAP@50:95 of 0.755. Although YOLO11n yielded the highest precision and mAP@50:95, YOLO26n attained the highest recall and mAP@50, indicating greater sensitivity and a lower likelihood of missed ROI detections. Given the sequential nature of the proposed framework, a localization failure at the ROI detection stage would propagate to the subsequent preprocessing and segmentation stages. Detection sensitivity was therefore prioritized over the marginal improvement in localization accuracy observed at stricter IoU thresholds.

To evaluate the contribution of ROI localization, direct full-frame segmentation was compared with the proposed ROI-based configuration. The segmentation performance of both configurations is summarized in Table 8.

Table 8: Comparison of full-frame and ROI-based lymphatic vessel segmentation.

Method	Precision	Recall	Dice	IoU	clDice
Full-frame YOLO26n-seg	0.867	0.847	0.8570	0.7500	0.820
ROI + YOLO26n-seg	**0.912**	**0.899**	**0.9050**	**0.8333**	**0.892**

As shown in Table 8, the ROI-based configuration achieved higher values for all evaluated metrics. Compared with direct full-frame segmentation, ROI-based segmentation increased precision from 0.867 to 0.912, recall from 0.847 to 0.899, and the Dice coefficient from 0.8570 to 0.9050. The IoU increased from 0.7500 to 0.8333, while clDice improved from 0.820 to 0.892. These results show that the ROI-based configuration was associated with improved pixel-level overlap and structural continuity under the evaluated conditions.

5.2. Effectiveness of Image Enhancement

To assess the contribution of the proposed preprocessing pipeline, both qualitative inspection and a controlled ablation study were conducted. The analysis examined whether image enhancement could improve the visibility of faint lymphatic vessels and provide more discriminative inputs for segmentation. To ensure a fair comparison, the YOLO26n-seg architecture, patient-level data partition, and training configuration were held constant across all experiments, with only the preprocessing strategy being varied.

Figure 5 portrays representative qualitative results illustrating the effects of the proposed image enhancement pipeline. Specifically, the original ICG images, WTH responses, contrast-stretched outputs, and corresponding ground-truth lymphatic vessel masks are compared. In the original images, several vessel segments are obscured by diffuse fluorescence and locally bright background regions. After WTH transformation, the slowly varying background responses are attenuated, while narrow bright structures become more prominent. Percentile-based contrast stretching further increases the intensity separation between the enhanced vessel responses and the residual background. Consequently, the processed images exhibit clearer vessel boundaries, improved structural continuity, and closer spatial agreement with the ground-truth masks.

Furthermore, Table 9 presents an ablation study of different preprocessing strategies for lymphatic vessel segmentation. Direct segmentation of the original ICG images (method #1) yielded a precision of 0.742, recall of 0.722, Dice coefficient of 0.7318, IoU of 0.6152, and clDice of 0.690. Median filtering alone (method #2) produced only modest improvements, indicating that noise suppression alone was insufficient to recover faint lymphatic structures. In contrast, WTH alone (method #3) increased the Dice coefficient to 0.8430 and the IoU to 0.7450. Contrast stretching alone (method #4) also improved segmentation

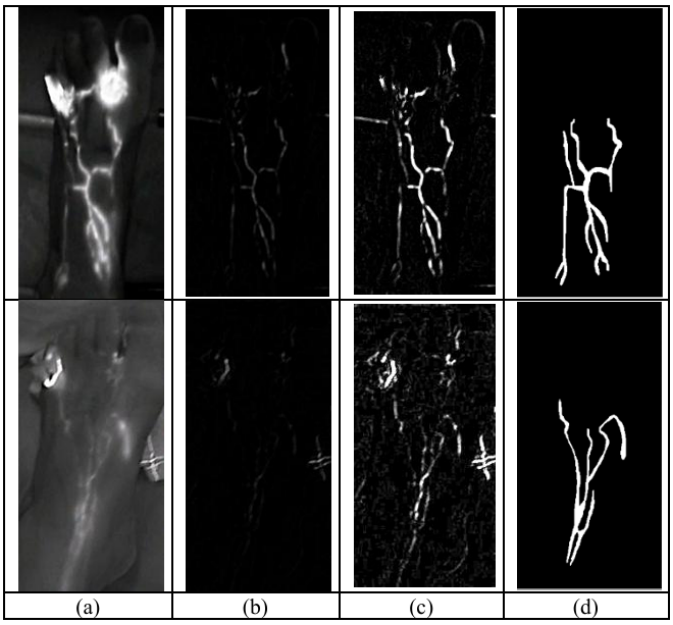


Figure 5: Representative qualitative results of the proposed image enhancement pipeline: (a) original ICG image, (b) WTH response, (c) percentile-based contrast stretching output, and (d) ground-truth lymphatic vessel mask.

performance, although the gains were smaller than those achieved using WTH. This comparison suggests that intensity enhancement without explicit background suppression is insufficient for accurately delineating lymphatic vessels under heterogeneous fluorescence conditions. The stronger performance of method #3 further highlights the importance of attenuating slowly varying background fluorescence.

Among the combined preprocessing strategies, WTH with contrast stretching (method #7) achieved a Dice coefficient of 0.8879, an IoU of 0.8048, and a clDice of 0.865. The complete pipeline (method #8), which integrates median filtering, WTH, and contrast stretching, achieved the best overall performance. Specifically, method #8 obtained a precision of 0.912, recall of 0.899, Dice coefficient of 0.9050, IoU of 0.8333, and clDice of 0.892. Compared with the original image setting (method #1), the proposed pipeline produced absolute improvements of 0.1732 in Dice, 0.2181 in IoU, and 0.202 in clDice. The increase in clDice indicates that the proposed pipeline improves not only regional overlap but also the topological continuity of the segmented lymphatic vessels.

The influence of the WTH structuring-element size was evaluated using four disk-shaped kernels. The corresponding segmentation results are presented in Table 10.

As shown in Table 10, the 9×9 kernel achieved the highest values for all evaluated metrics, with a precision of 0.912, recall of 0.899, Dice coefficient of 0.9050, IoU of 0.8333, and clDice of 0.892. The 5×5 kernel produced the lowest performance, whereas the results progressively improved with the 7×7 and 9×9 kernels. Increasing the kernel size to 11×11 resulted in a slight decrease in performance. These findings indicate that the 9×9 kernel provides the most favorable spatial scale for WTH-based lymphatic vessel enhancement among the evaluated configurations.

5.3. Five-Fold Cross-Validation

To further evaluate the robustness of the proposed framework, five-fold cross-validation was conducted using patient-level partitioning. In each fold, patients were divided so that frames from the same patient did not appear simultaneously in the training and validation subsets. This design reduced the risk of patient-specific information leakage and provided a more reliable estimate of generalization performance.

As shown in Table 11, the proposed framework achieved stable performance across all folds, with mean Dice, IoU, precision, recall, and clDice values of 0.903, 0.829, 0.909, 0.897, and 0.889, respectively. The small standard deviations across folds indicate that the framework maintained consistent segmentation performance under different patient-level data partitions. The narrow performance ranges across the five folds, with Dice values from 0.896 to 0.912 and clDice values from 0.881 to 0.898, further suggest limited sensitivity to variations in cohort composition. Moreover, the close agreement between the Dice and clDice results indicates that the framework preserves both regional overlap and the topological continuity of the segmented lymphatic vessels.

5.4. Comparison of Lightweight Segmentation Models

After establishing the effectiveness of the proposed image enhancement pipeline, four lightweight segmentation models were evaluated under a unified experimental protocol, as summarized in Table 12. The comparison included Fast-SCNN [35], BiSeNetV2 [36], YOLO11n-seg [34], and YOLO26n-seg [26] model.

As shown in Table 12, YOLO26n-seg (method #4) achieved the best segmentation performance, with a Dice coefficient of 0.9050 and an IoU of 0.8333. Compared with Fast-SCNN (method #1), method #4 improved the Dice coefficient by 0.0526 and the IoU by 0.0865. It also outperformed YOLO11n-seg (method #3) by 0.0314 in Dice and 0.0361 in IoU. These improvements indicate that YOLO26n-seg provides more accurate delineation of faint and spatially discontinuous lymphatic vessels.

A clear tradeoff between segmentation performance and computational efficiency was observed across the evaluated architectures. Fast-SCNN (method #1) contained the fewest parameters, but its lower Dice and IoU values suggest that its highly compressed feature representation was less effective in preserving fine vessel boundaries. BiSeNetV2 (method #2) achieved higher segmentation accuracy through its dual path architecture, although it required more parameters than the other models. YOLO11n-seg (method #3) achieved stronger over-

Table 9: Ablation study of different preprocessing strategies for lymphatic vessel segmentation. Filt. denotes filtering, whereas str. denotes stretching.

No.	Preprocessing strategy	Precision	Recall	Dice	IoU	clDice
1	Original image	0.742	0.722	0.7318	0.6152	0.690
2	Median filtering only	0.765	0.748	0.7560	0.6350	0.718
3	WTH only	0.852	0.835	0.8430	0.7450	0.816
4	Contrast str. only	0.802	0.772	0.7864	0.6747	0.760
5	Median filt. + WTH	0.871	0.852	0.8610	0.7790	0.840
6	Median filt. + contrast str.	0.820	0.796	0.8076	0.6973	0.784
7	WTH + contrast str.	0.894	0.882	0.8879	0.8048	0.865
8	**Proposed pipeline (median filt. + WTH + contrast str.)**	**0.912**	**0.899**	**0.9050**	**0.8333**	**0.892**

Table 10: Sensitivity analysis of the WTH structuring-element size.

WTH kernel	Precision	Recall	Dice	IoU	clDice
5×5	0.856	0.844	0.8499	0.7354	0.808
7×7	0.894	0.878	0.8858	0.7997	0.859
9×9	**0.912**	**0.899**	**0.9050**	**0.8333**	**0.892**
11×11	0.901	0.888	0.8944	0.8159	0.872

lap performance than Methods #1 and #2, demonstrating the suitability of YOLO based mask prediction for this application. Nevertheless, YOLO26n-seg (method #4) achieved the highest accuracy while requiring only 2.6 million parameters and an average inference time of 3.5 ms.

The superior performance of method #4 can be attributed to its integration of multiscale feature extraction, efficient feature aggregation, and mask decoding. Fine lymphatic vessels exhibit narrow boundaries, weak local contrast, and substantial variations in spatial scale. Preserving these structures requires both high resolution spatial information and sufficient contextual representation. YOLO26n-seg maintains this balance without introducing excessive computational complexity, thereby improving vessel continuity and reducing missed vessel segments.

5.5. Visual Assessment of Segmentation

To complement the quantitative evaluation, representative segmentation results are presented in Figure 6. The figure compares the original ICG images, clinician-annotated vessel masks, and predictions generated by YOLO26n-seg. This qualitative assessment focuses on the spatial agreement of the vessel trajectories, the preservation of branching structures, and the suppression of non-vessel background responses.

As shown in Figure 6, the predicted masks generally follow the principal lymphatic pathways identified in the ground-truth annotations. The model captures the dominant vessel trunks and preserves several clinically relevant branching patterns. Most diffuse fluorescence regions are excluded from the predicted masks, indicating that the model can distinguish vessel-like structures from heterogeneous background responses. The predicted vessel trajectories also remain spatially continuous across regions with moderate fluorescence contrast.

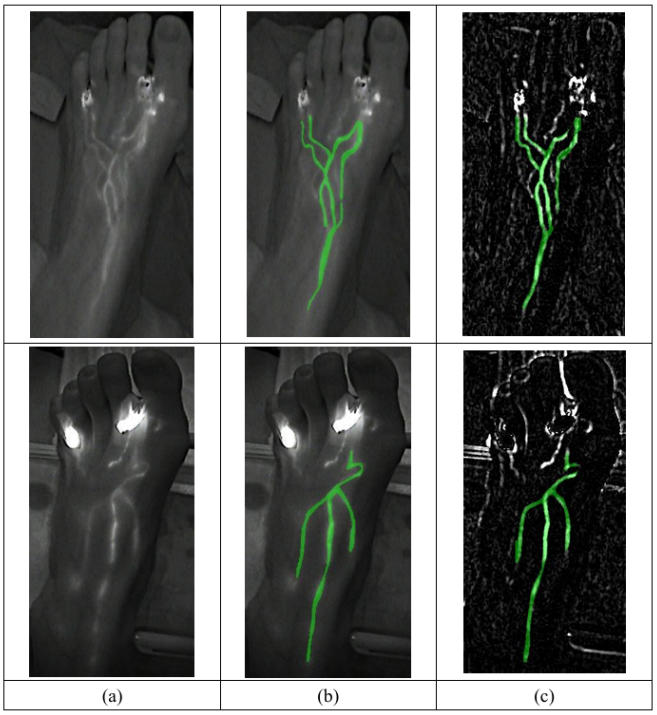


Figure 6: Representative YOLO26n-seg results: (a) original ICG image, (b) ground-truth overlay, and (c) predicted-mask overlay.

Minor discrepancies are observed in weakly fluorescent and poorly defined vessel segments. Some narrow peripheral branches are incompletely delineated, whereas several locally enhanced structures produce small false-positive responses. These differences are mainly associated with low vessel-to-background contrast, diffuse dermal reflux, and ambiguous vessel boundaries. Under these conditions, the local intensity and morphological characteristics of lymphatic vessels may overlap with those of surrounding fluorescence artifacts, thereby increasing segmentation uncertainty.

5.6. Frame-wise Processing Performance on ICG Video Sequences

The frame-wise processing performance of the proposed pipeline was evaluated using four ICG lymphography video sequences. After image-based training, YOLO26n-seg was

Table 11: Five-fold cross-validation performance of the proposed framework.

Fold	Dice	IoU	Precision	Recall	clDice
1	0.905	0.833	0.912	0.899	0.892
2	0.899	0.824	0.905	0.893	0.886
3	0.912	0.839	0.918	0.906	0.898
4	0.896	0.820	0.901	0.891	0.881
5	0.904	0.831	0.910	0.898	0.890
Mean ± SD	0.903 ± 0.006	0.829 ± 0.008	0.909 ± 0.006	0.897 ± 0.006	0.889 ± 0.006

Table 12: Performance and computational efficiency of the evaluated lightweight segmentation models. Inf. denotes inference time, while Params. denotes the number of parameters.

	Model	Dice	IoU	Params. (M)	Inf. time (ms)
1	Fast-SCNN [35]	0.8524	0.7468	1.2	4.4
2	BiSeNetV2 [36]	0.8679	0.7675	3.4	4.2
3	YOLO11n-seg [34]	0.8736	0.7972	2.9	4.0
4	**YOLO26n-seg** [26]	**0.9050**	**0.8333**	**2.6**	**3.5**

Table 13: Frame-wise processing performance across clinical ICG lymphography sequences

Video	Frame #	Source FPS	Core time (ms/frame)	Max.throughput FPS
1	126	24.0	19.4	29.8
2	117	30.0	18.4	30.6
3	115	30.0	19.1	30.5
4	121	30.0	19.3	30.7

independently applied to each frame. Every frame underwent ROI localization, spatial standardization, image enhancement, lymphatic vessel segmentation, mask remapping, and overlay generation. No temporal information was exchanged between consecutive frames. Core processing time and the maximum unpaced processing throughput of the complete pipeline were measured, as summarized in Table 13.

In this evaluation, source FPS denotes the native acquisition and playback frame rate of each clinical video. All evaluated sequences had a spatial resolution of 540×960 pixels. Core processing time represents the average time required for ROI localization, spatial standardization, image preprocessing, and YOLO26n-seg inference. Maximum processing throughput additionally includes video decoding, frame acquisition, mask remapping, overlay generation, and output encoding. It was measured under unpaced decoding conditions and therefore represents the computational capacity of the complete pipeline rather than the actual output video frame rate.

As shown in Table 13, the core processing time ranged from 18.4 to 19.4 ms per frame, corresponding to a core computational capacity of approximately 51.5–54.3 FPS. After video input/output operations and overlay rendering were included, the maximum unpaced processing throughput ranged from 29.8 to 30.7 FPS, with an average of 30.4 FPS. During normal video output, the presentation rate remains limited by the source frame rate of 24 or 30 FPS.

The difference between the core processing speed and the end-to-end throughput is mainly attributable to frame decoding, memory transfer, mask remapping, visualization, and output encoding. These operations introduce additional overhead that is not reflected in model-only inference measurements. The measured maximum processing throughput was approximately equal to or higher than the source frame rates of the evaluated sequences. This result indicates that the pipeline had sufficient computational capacity to process incoming frames without substantial backlog under the evaluated hardware conditions.

6. Discussion

This section discusses the findings of the proposed framework and their implications for real-time lymphatic vessel visualization. Specifically, the effects of the image-enhancement pipeline and the performance of the selected lightweight segmentation model are discussed in terms of segmentation accuracy and computational efficiency. The section then concludes by outlining the main limitations of the current work, which include restricted frame sampling, reliance on a single-center dataset, limited anatomical coverage, and the lack of prospective clinical validation.

6.1. Main Findings

The proposed framework substantially improved lymphatic vessel segmentation in low-contrast ICG lymphography. Compared with direct segmentation of the original images, the complete preprocessing pipeline increased the Dice coefficient from 0.7318 to 0.9050 and the IoU from 0.6152 to 0.8333. These findings indicate that segmentation performance was driven not only by the network architecture, but also by the quality of the input representation. In particular, suppressing nonuniform fluorescence and enhancing faint tubular structures reduced interference from dermal reflux, tissue scattering, and background fluorescence.

The ablation results further demonstrated the complementary roles of the preprocessing stages. Median filtering reduced localized fluorescence noise while preserving vessel boundaries. The WTH transformation produced

the largest performance gain by attenuating slowly varying background responses and emphasizing narrow vessel structures. Percentile-based contrast stretching then increased the visibility of the remaining vessel responses. Together, these operations provided YOLO26n-seg with a cleaner and more discriminative representation of lymphatic morphology.

The lightweight segmentation model also achieved a favorable balance between accuracy and computational efficiency. YOLO26n-seg obtained the highest Dice and IoU values among the evaluated models while requiring only 2.6 million parameters and an average inference time of 3.5 ms. In the frame-wise sequence evaluation, the complete pipeline achieved a maximum unpaced processing throughput of approximately 30.4 FPS. This performance suggests that ROI localization, image enhancement, segmentation, and mask visualization can be integrated without disrupting real-time video playback. The resulting overlays may therefore support continuous visualization of candidate lymphatic pathways during LVA-related procedures.

6.2. Limitations

Although the proposed pipeline achieved favorable segmentation accuracy and processing efficiency, its current evaluation remains preliminary. First, only three diagnostically usable and reliably annotatable frames were manually selected from each of the 164 patients. This selection strategy may introduce sampling bias and may not fully represent the complete range of fluorescence changes, motion artifacts, vessel visibility, and dermal reflux observed throughout the original ICG video sequences. Consequently, the reported performance should be interpreted as the segmentation performance on selected annotatable frames rather than on all frames from continuous clinical videos. Future studies should include denser frame sampling and video-level annotation across different phases of lymphatic flow.

The current study primarily demonstrates technical feasibility based on segmentation accuracy and processing efficiency. Its effect on clinical decision-making, anastomosis site selection, procedure duration, and patient outcomes remains to be established. Prospective clinical studies should compare conventional visual assessment with AI-assisted guidance under real surgical conditions. Annotation uncertainty is another potential source of bias because faint vessel boundaries may be interpreted differently by individual clinicians. This issue could be mitigated through independent annotations by multiple microsurgeons, consensus review, and interobserver agreement analysis. In addition, the reported real-time performance was measured on a workstation equipped with GPU acceleration. Further optimization, model quantization, and hardware-specific benchmarking are therefore required before deployment on compact edge devices or integrated surgical platforms.

7. Conclusion

This work presents a coarse-to-fine, low-latency pipeline for frame-wise lymphatic vessel segmentation in near-infrared indocyanine green lymphography. The proposed method combines clinically focused region localization with image enhancement and lightweight instance segmentation to address uneven fluorescence backgrounds, dermal reflux, tissue scattering, and low vessel-to-background contrast. On the evaluated dataset, the complete preprocessing pipeline increased the Dice coefficient from 0.7318 to 0.9050 and the IoU from 0.6152 to 0.8333 while maintaining low processing latency.

The pipeline first uses a lightweight detection network to localize the affected foot region in each frame. Aspect-ratio-preserving resizing and symmetric zero padding are then applied to standardize the input dimensions without substantially distorting the anatomical structures. Median filtering, morphological white top-hat transformation, and percentile-based contrast stretching are subsequently used to reduce background variation and improve the visibility of faint lymphatic vessels. Finally, YOLO26n-seg generates pixel-level vessel masks, which are mapped back onto the corresponding original frames for sequential visualization.

Overall, the proposed pipeline achieved a favorable balance between segmentation accuracy and computational efficiency under the evaluated dataset and hardware conditions. By integrating ROI localization, image enhancement, lightweight segmentation, and mask visualization, the framework provides a practical technical basis for computer-assisted analysis of ICG lymphography. Further evaluation using larger multicenter datasets and prospective clinical studies will be necessary to determine its generalizability and potential value in intraoperative applications.

Acknowledgments

This work was funded by National Science and Technology Council (NSTC), R.O.C (Grant Number: NSTC 114-2221-E-035-006- and NSTC 114-2221-E-035-020-MY3).

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